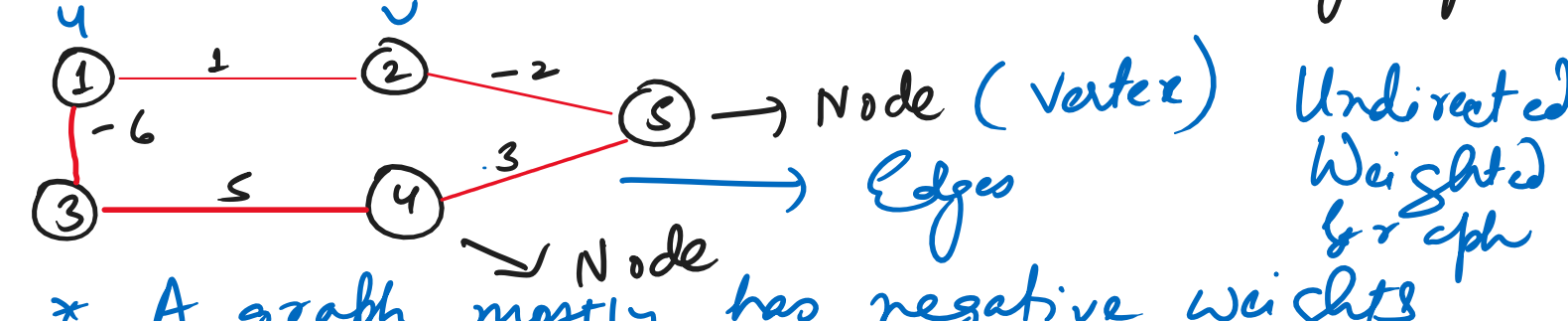
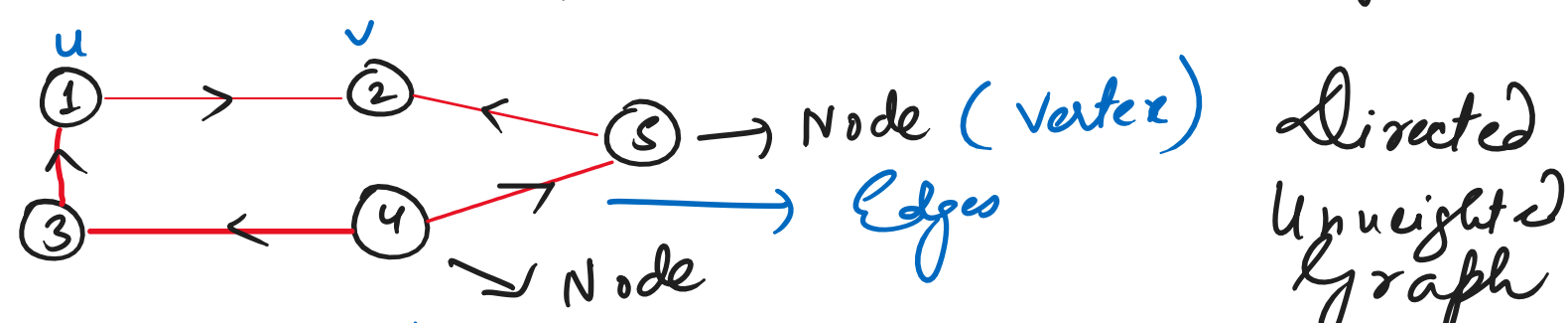
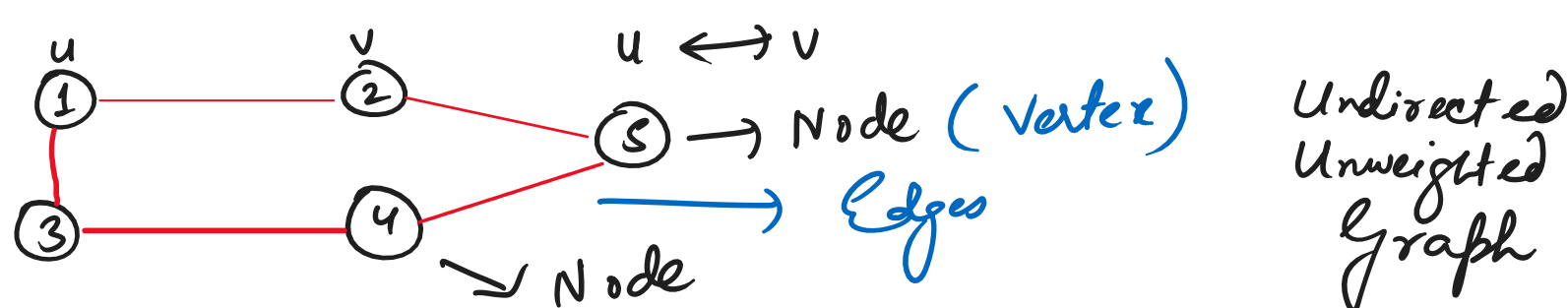


* Non-linear Data Structures Continued!

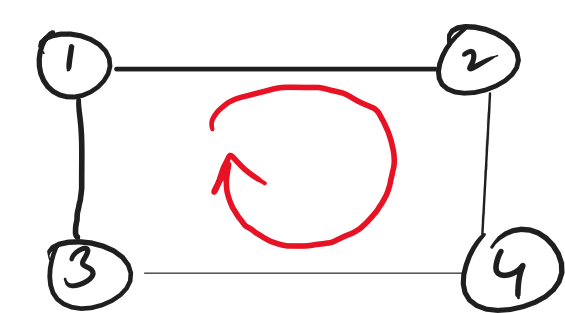
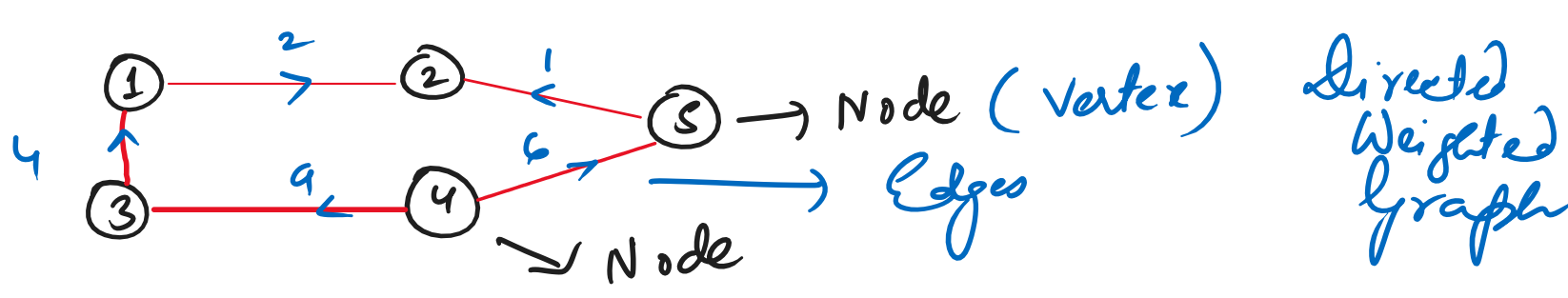
* Introduction to the Graph Data Structure :->

Definition :-> A graph is a non-linear data structure which contains entities called nodes or vertices which are connected to each other via edges.

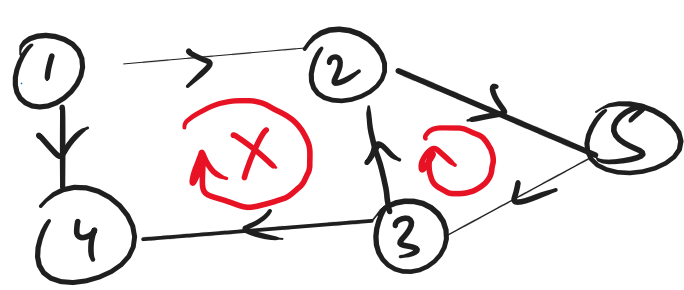
Depending on the type of connectivity, there can be various types of graphs :->



* A graph mostly has negative weights on maybe performance, sales, etc.

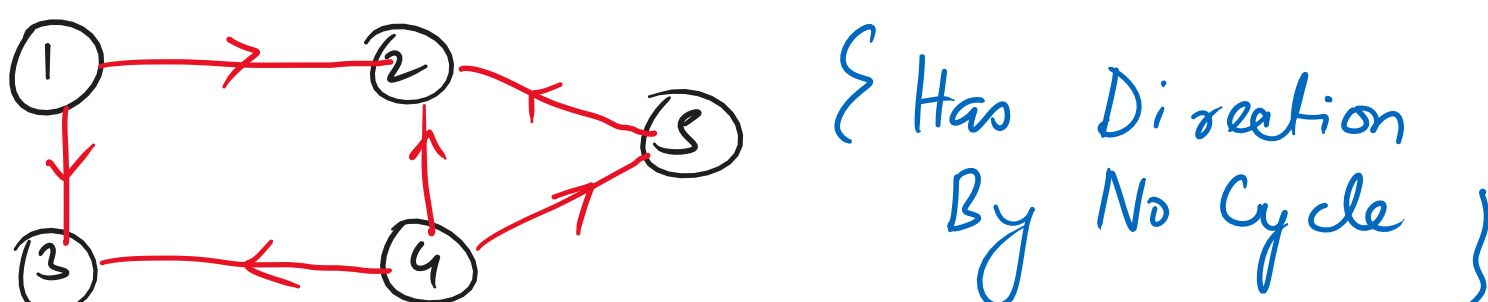


From any node we can cover the graph & come back to itself.

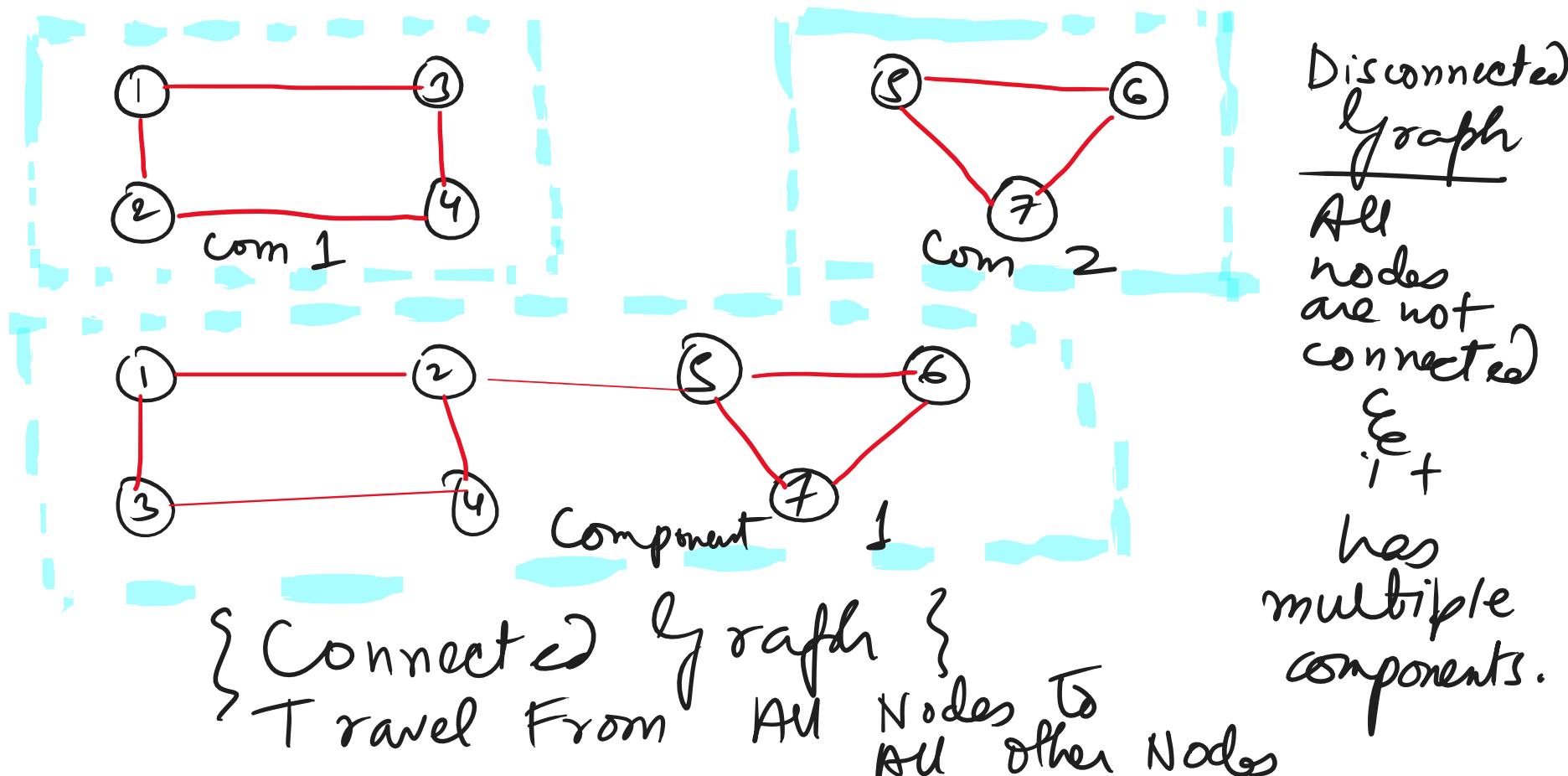


* DAG : (Directed Acyclic Graph)

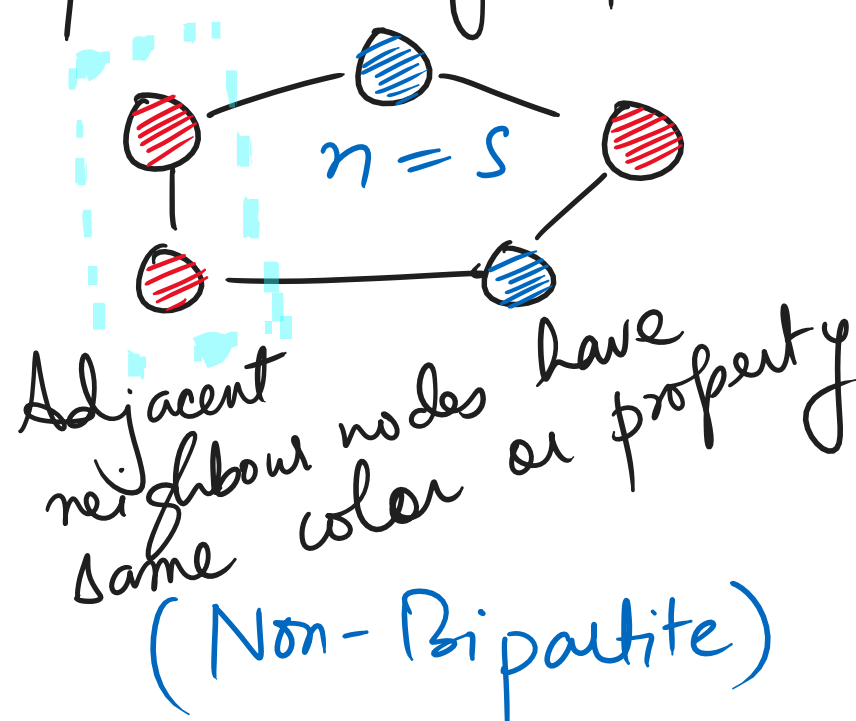
** Interview (Topological Sort) (DFS, BFS, Kahn's Algorithm)



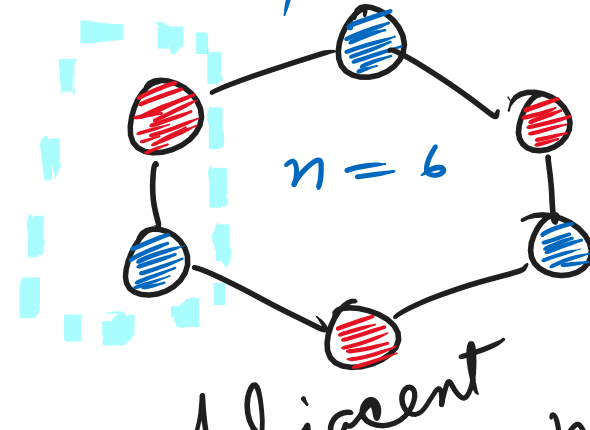
* Cycle Detection -> Topo Sort, BFS, DFS



Bipartite Graph :->



(Bipartite)



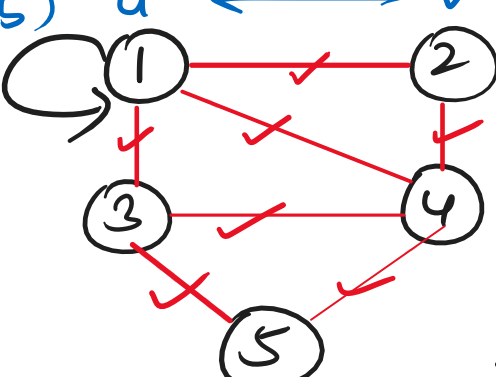
* All even connected graphs are bipartite

Graph Representations :-> Time Complexity

Adjacency Matrix

	1	2	3	4	5
1	0	1	1	1	0
2	1	0	0	1	0
3	1	0	0	1	1
4	1	1	1	0	1
5	0	0	1	1	0

Node to Self is considered as 0 or invalid



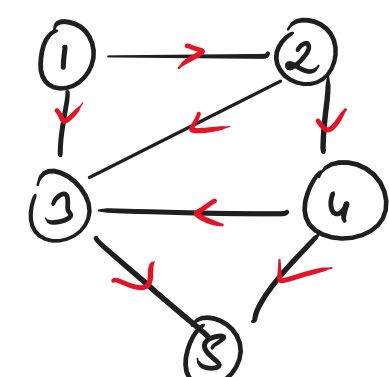
$O(V)$
 $O(n)$ because we traverse all n nodes.

Adjacency List

Node : List of Neighbours
 $i = 1 : \{2, 3, 4\}$
 $i = 2 : \{1, 4\}$
 $i = 3 : \{1, 4, 5\}$
 $i = 4 : \{1, 2, 3, 5\}$
 $i = 5 : \{3, 4\}$

Adjacency Matrix

	1	2	3	4	5
1	0	1	1	0	0
2	0	0	1	1	0
3	0	0	0	0	1
4	0	0	1	0	1
5	0	0	0	0	0

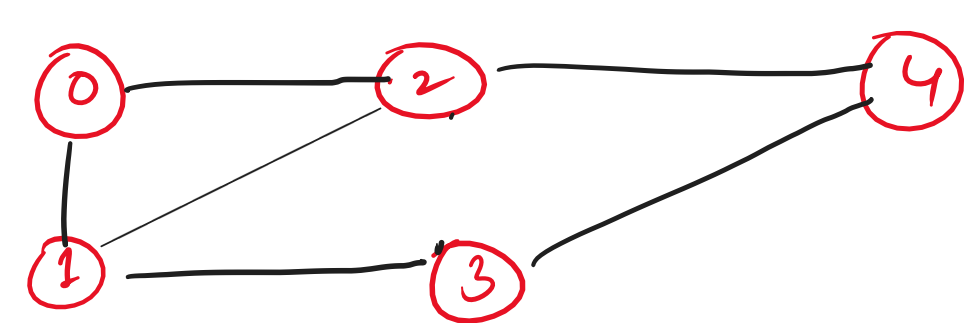


Adjacency List

Node : List of Neighbours
 $1 : \rightarrow 2, 3$
 $2 : \rightarrow 3, 4$
 $3 : \rightarrow 5$
 $4 : \rightarrow 3, 5$
 $5 : \rightarrow$

Graph Nodes & Edges Counting :->

nodes
adj list.size();



no of edges 12 / 12 / 2 = 6 edges